



Netscore

NetScore

October 2024 — Ongoing

A scoreboard system designed to track game scores in real time using one or more Bluetooth buttons. Each button can be assigned to increase the home or away score, making it easy to update the display without touching the main unit.

The system runs on an ESP32-S2 and drives six 7-segment displays through **TLC5926** constant-current drivers, showing both points and sets for home and away teams. The 7-segments displays are **multiplexed in pairs** and duplicated on a **mirrored back board**, so the score is visible from both sides of the court.

Originally, the plan was to increment the score directly on the device itself — it's mounted on the net, so anyone nearby could press a button to update it. After some feedback from friends, it became clear that external BLE buttons would be a better approach. That's when I hit a snag: the ESP32-S2 doesn't have Bluetooth.

Since my hardware was already designed around the S2, I had to improvise. The current prototype uses an ESP32-S2 as the main controller and a separate ESP32 as a "Bluetooth hub." The two communicate via ESP-NOW, with the ESP32 handling all Bluetooth input from the legacy BLE buttons.

Some of the structure and ideas came from my earlier **Goalie Rotation Timer** prototypes, and future versions will likely merge both concepts — scoring and timed rotations — into one system.

All source files are available on [GitHub](#).

Notes:

- ESP32-S2 main controller + ESP32 Bluetooth hub
- Communication handled via ESP-NOW
- BL button input for score updates

- TLC5926 drivers controlling **six multiplexed 7-segment displays**
- **Mirrored display setup** (front + back)
- Inspired by the earlier **Goalie Rotation Timer** prototypes
- Future plans include integrating rotation timing and other game modes

Architecture:

The system is split between two microcontrollers. The **ESP32-S2** acts as the main controller, managing the display logic, TLC5926 drivers, and overall state of the scoreboard. The **ESP32** functions as a Bluetooth hub, receiving input from legacy BLE buttons and forwarding the events to the S2 via **ESP-NOW**.

This setup allowed development to continue on existing S2-based hardware while maintaining BLE functionality through an external board. Future revisions will likely consolidate everything into a single BLE-capable microcontroller.

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October 2024 — Ongoing / Embedded Systems, IoT

A real-time, dual-sided scoreboard system that uses external Bluetooth buttons for control and score updates.

(This is the perfect place for photos of the scoreboard in action, the internal electronics, and perhaps a short video demonstrating the BLE button functionality.)

The Story

NetScore is a scoreboard system designed to track game scores in real time using one or more Bluetooth buttons. Each button can be assigned to increase the home or away score, making it easy for players or referees to update the display wirelessly. The system runs (for now) on an ESP32-S2 and drives 3 pairs of multiplexed 7-segment displays, showing both points and sets. To ensure the score is visible from both sides of the court, the displays are mirrored on a back-to-back board.

Originally, the plan was to increment the score directly on the device itself - it's mounted on the net, so anyone nearby could press a button to update it. After

some feedback from friends, it became clear that external BLE buttons would be a better approach. That's when I hit a snag: the ESP32-S2 lacks native Bluetooth. Instead of starting over, I added a separate ESP32 to act as a "Bluetooth hub." This ESP32 handles all BLE input and forwards events to the ESP32-S2 controller via ESP-NOW, a proprietary peer-to-peer communication protocol. Some of the structural ideas were inherited from my earlier Goalie Rotation Timer, and I plan to merge both concepts - scoring and timed rotations, for football - in a future version.

Key Features

- Real-time score updates via Bluetooth Classic buttons.
- Displays both points and sets for home and away teams.
- Six multiplexed 7-segment displays controlled by TLC5926 constant-current drivers.
- Unique dual-MCU architecture communicating via ESP-NOW.

Tech Stack

- **Main Controller:** ESP32-S2
- **Bluetooth Hub:** ESP32
- **Wireless Protocols:** ESP-NOW, Bluetooth Classic
- **Display Drivers:** TLC5926 Constant-Current LED Drivers
- **Programming:** C/C++ (ESP-IDF)

Challenges & Lessons Learned

The primary challenge was discovering that my chosen main controller, the ESP32-S2, did not support Bluetooth after the hardware was already designed. This led to a creative workaround: implementing a two-chip solution. This experience was a valuable lesson in hardware specification and adaptability, forcing me to learn and implement ESP-NOW as a robust communication bridge between two microcontrollers. It demonstrated how to salvage a design by augmenting it rather than starting from scratch.

Links

- <https://github.com/arcticlula/Netscore>

Ideas

DS3231 rtc

tlc